

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (EC) Sem -III Examination (Backlog), 2010-11

EE220 Electrical Machines & Power Circuits

Date: 21/05/11, Saturday

Time: 10:00 am to 01:00 pm

Maximum Marks:70

Instructions:

- [1] Attempt all the questions.
- [2] Figures to the right indicate full marks.
- [3] Assume suitable data wherever necessary.

SECTION- I

- Q1(A) Explain the $N \rightarrow I_a$ & $T \rightarrow I_a$ characteristics for D C series and shunt motor. 05
- (B) A long shunt generator gives full load output of 20kW at a terminal voltage of 220 V. The armature resistance, series field and shunt field resistances are 0.04Ω , 0.05Ω and 110Ω respectively. Overall efficiency at this load is 85%. Find: 05
- (1) Cu Losses
 - (2) Iron and friction losses
 - (3) Torque developed by the prime mover

OR

- Q1(A) Explain the voltage build up process of a self-excited DC generator. 05
- (B) Explain any one method of speed control of DC shunt Motor. 05
- Q2(A) Explain the functioning of transformer at load. 04
- (B) The following readings are obtained for O.C. and S.C. tests performed on 200/400 volt, 50 Hz, 1- Φ transformer, 06
- O.C. test: 200 V, 0.7 A, 70 W.
- S.C. test: 15 V, 10 A, 85 W.
- Compute:
- (1) Equivalent circuit parameters
 - (2) The secondary voltage when delivering 5 kW at 0.8 pf lagging with the primary voltage of 200 V.

OR

- (B) Answer the following questions: 06
- (1) Derive the condition for the maximum efficiency in single-phase transformer.
 - (2) Explain the difference between the core type and shell type of transformer
- Q3 Write a short note on the following [Any *Three*] 15
- (1) List the equipments used in the sub-station with their application.
 - (2) Internal and external characteristics of a shunt generator
 - (3) Conditions for parallel operation of two single phase transformer
 - (4) Comparison of indoor substation with outdoor substation
 - (5) DC shunt motor starter [3-point]

SECTION – II

- Q4(A) Derive the torque expressions for the 3- Φ induction motor and obtain the condition for maximum torque. 05
- (B) Explain the double field revolving theory for single-phase induction motor. 05
- OR**
- Q4(A) Explain the operation of single-phase induction motor with the help of split phase technique. 05
- (B) For a 3-phase, 8 pole slip ring induction motor has full load slip of 4%. The rotor impedance per phase is given as $(0.01+j0.1) \Omega$. Find the ratio of maximum to full load torque and the speed at which the maximum torque occurs. 05
- Q5(A) Derive the EMF equation of an alternator. 04
- (B) A 3- ϕ , 16-pole synchronous generator has a resultant air gap flux of 0.05 Wb/pole. The flux distributed sinusoidally over the pole. The stator has 144 slots and 4 conductors per slot accommodated in two layers. The coil span is 150° electrical. Calculate the phase and line induced voltages when the machine runs at 375 rpm. 06
- OR**
- Q5(A) Explain why armature kept stationary and field rotating in synchronous machine. 04
- (B) A 3-phase, star connected alternator is rated at 500 kVA, 1.1 kV. The armature resistance and synchronous reactance are $0.1 \Omega/\text{phase}$ and $1.5\Omega/\text{phase}$ respectively. Calculate its voltage and %regulation for
- (1) Unity power factor
 - (2) 0.9 lagging power factor
 - (3) 0.8 leading power factor
- Q6 Write a short note on the following [Any *Three*] 15
- (1) Torque slip characteristic of a 3-phase induction motor
 - (2) Methods for power factor improvement
 - (3) Desirable characteristics of a Tariff
 - (4) Shaded pole single phase induction motor
 - (5) Distribution factor of alternator
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